Shahrzad Dehghani

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Solingen, NRW, Germany Dual Citizen: German, Iranian
Portfolio LinkedIn ORCiD GitHub ResearchGate



Profile

Machine Learning Engineer with 3+ years of developing & benchmarking ML & deep learning solutions for large-scale scientific & industrial datasets. Skilled in Python, SQL, PyTorch, Databricks, Apache Spark, & HPC environments, with end-to-end ML experience spanning data preparation, model development, hyperparameter optimization, robustness evaluation, and deployment. Complemented by independent data analytics, NLP and LLM experience, including the analysis of 6.8M Amazon reviews using statistical analysis, machine learning & transformer fine-tuning to extract insights into customer sentiment & product performance.

Skills

- **Programming Languages & Tools:** Python (Pandas, NumPy, SciPy, etc.), SQL, MATLAB, C++
- **Machine Learning & Deep Learning:** PyTorch, Scikit-learn, TensorFlow, Keras, CNNs, Generative models (cVAE, cGAN, etc), Hyperparameter Optimization (Optuna), Model Benchmarking & Robustness Evaluation
- **NLP, LLMs & Agentic AI:** NLTK, Sentiment Analysis, Hugging Face Transformers, Transformer Fine-tuning (DistilBERT, RoBERTa), LLM Agents, Agentic Workflows, Tool/Function Calling
- **Data Engineering & Platforms:** Databricks, Apache Spark/PySpark, Delta Lake, ETL/ELT Pipelines, Medallion Architecture, Unity Catalog, Delta Live Tables (Lakeflow), Databricks Jobs, Data Warehousing & Lakehouse Architecture
- **Data Analysis & Statistics:** Data Wrangling, Exploratory Data Analysis (EDA), Regression & Statistical Analysis, KPI Development, A/B Testing Concepts
- **Computer Vision & Imaging:** OpenCV, Hyperspectral Image Analysis, Image Processing
- **Data Visualization:** Power BI, Tableau, Databricks AI/BI Dashboards, Matplotlib, Seaborn, Excel
- **High-Performance Computing (HPC):** Distributed CPU/GPU Computing, Job Scheduling (SLURM), Parallel Data Processing
- **Version Control:** Git, GitHub
- **Software & Application Development:** HTML, CSS, Python GUI Development

Supporting courses, certificates, and practical projects are available in my Portfolio.

Employment History

- **Machine Learning Researcher & Engineer (Researcher in Optoelectrical Engineering) · Bergische Universität Wuppertal, Germany** *08.2023 – Present*

  - Developed physics-based simulations using Rigorous Coupled-Wave Analysis (RCWA) to generate 15,000 reflection spectra for nanophotonic structures. Developed and benchmarked deep learning architectures, including discriminative and generative AI architectures, to solve an inverse design problem. Predicted structural parameters with a mean absolute error below 1 nm for both silver layer thickness and grating period. The models were trained and evaluated for robustness against measurement noise. Implemented in Python/PyTorch with hyperparameter optimization (Optuna) and deployed on HPC environments using SLURM. Here are project explanation and paper.
  - Applied machine learning and data science techniques in a confidential industry collaboration with Louisenthal GmbH, including data preparation, predictive modeling, model optimization, and quantitative performance analysis.
  - Developed a Python based GUI application for the evaluation of university applicants; the software is currently in active use by the Bergische Universität Wuppertal.
  - Teaching & supervision: tutorial sessions in thin film technology, including explaining theoretical concepts and guiding practical exercises. Supervised bachelor thesis project and seminar presentations for undergraduate students.

Employment History (continued)

Laboratory based research internship in experimental physics · Universität zu Köln *11.2022 – 07.2023*

- Conducted thin-film experiments under ultra-high vacuum (UHV) conditions and analyzed spectroscopy datasets (XPS, UPS, IPES, REELS) using Python and OriginLab.
- Performed data preprocessing, cleaning, analysis, and visualization of experimental data to extract scientific insights.
- Developed and optimized automation scripts for data processing, analysis workflows, and experimental evaluation, improving efficiency and reproducibility.

Research Assistant (Data Analysis & HPC — Master Thesis) · Forschungszentrum Jülich *9.2021 – 08.2022*

- Simulated, processed, and analyzed large-scale datasets from N-Body6++ star cluster models at the Jülich Supercomputing Centre using Python, C++, and HPC systems (SLURM). Managed the complete data pipeline, including data cleaning, processing, and visualization.
- Developed multi-scenario comparative statistical analyses across different model parameters to quantify environmental effects on protoplanetary disc sizes (Master thesis).
- Contributed to an independent research collaboration on the statistical analysis of observational data concerning the lifetime of protoplanetary disks; results were published in a peer-reviewed scientific publication (Paper).

Summer Internship · Universität Bonn, Germany *7.2020 – 10.2020*

- Analyzed hyperspectral image datasets using Python (SPy) to extract physical and chemical properties from multi-dimensional data.

Research Assistant · Institute for Research in Fundamental Sciences (IPM) in Tehran, Iran *02.2018 – 12.2019*

- Processed and analyzed astronomical observational datasets to create scientific stellar catalogs.
- Performed data cleaning, quality control, and validation of photometric measurements obtained from INT telescope observations.
- Applied statistical analysis techniques to identify variable stars and investigate the star formation history of galaxies.
- Integrated and cross matched multiple data sources, including telescope observations, Gaia catalogs, and external surveys, to improve data quality and completeness.
- Contributed to research projects that resulted in multiple scientific publications (Paper 1 and Paper 2).

Independent Projects

Mining insights from Amazon Jewelry Reviews *03.2026 – Present*

- From business analytics to insight generation using LLMs (Python, Power BI, SQL)
- **Goal**: Analyzing Amazon jewelry reviews to uncover customer satisfaction drivers, brand performance, and product issues using both data analytics and NLP/LLM techniques.
- **Part 1: Data Analysis & Business insights: brand benchmarking, price-segment analysis, and review-text mining.**
- Cleaned and wrangled a large scale dataset (6.8M reviews, 800k products, 23k brands) spanning 19 years of Amazon jewelry reviews, handling missing values, text normalization, and schema inconsistencies.
- Conducted EDA and NLP based review text analysis on Amazon jewelry dataset to uncover insights on product trends, brand performance, pricing, ratings, and customer sentiment.
- Identified key drivers of customer dissatisfaction by applying regression analysis. The analysis revealed that complaints related to durability and appearance had the strongest negative impact on product ratings across all price segments.

Independent Projects (continued)

- Built an interactive Power BI dashboard visualizing brand KPIs, pricing trends, and category performance across 19 years of data.
- **Part 2: Data Science (NLP & Transformer-based modeling)**
- Developed a transformer-based sentiment classification on review data. Fine-tuned DistilBERT and RoBERTa (Hugging Face) for 3-class sentiment classification; achieved 93% accuracy and macro F1 0.80 on held-out test data.
- Extending the project with topic modeling, aspect-based sentiment analysis, and fake review detection.
- The full analysis, codes, results, visualizations, and thorough explanations are available in my Github and my Portfolio.

Education

M.Sc. in Physics, Universität zu Köln *04.2020 – 03.2023*
Thesis title: *Role of gas expulsion time on cluster expansion. [Jülich Supercomputing Center (JSC)]*

M.Sc. in Theoretical Physics, Alzahra University, Tehran, Iran *10.2014 – 07.2016*
Full scholarship as a top student in the national entrance exam for master's programs

B.Sc. in Physics, Shahid Beheshti University, Tehran, Iran *10.2009 – 08.2014*
Specialization : Atomic and Molecular Physics - Full scholarship as a top 1% student in the national Iranian university entrance exam.

Research Publications

1. S. Dehghani, C. Knoth, et al., "Data-driven inverse design of hybrid waveguide gratings using reflection spectra via tandem networks and conditional vaes," *MDPI - Optics*, Nov. 2025. DOI: 10.3390/opt6040061.
2. T. Parto, S. Dehghani, et al., "The isaac newton telescope monitoring survey of local group dwarf galaxies. v. the star formation history of sagittarius dwarf irregular galaxy derived from long-period variable stars," *The Astrophysical Journal*, Jan. 2023. DOI: 10.3847/1538-4357/aca471.
3. S. Pfalzner, S. Dehghani, et al., "Most planets might have more than 5 myr of time to form," *The Astrophysical Journal Letters*, Oct. 2022. DOI: 10.3847/2041-8213/ac9839.
4. T. Parto, S. Dehghani, et al., "Int monitoring survey of local group dwarf galaxies: Star formation history and chemical enrichment," *Communications of the Byurakan Astrophysical Observatory*, 2020, ISSN: 2579-2776. DOI: 10.52526/25792776-2020.67.2-232.

Languages

Languages: English (C1), German (B2 – Improving), Persian (C2).

Volunteering

Build Your Future *06.2017 - 08.2017*
Held Soft Skills Workshops (AIESEC in Baku, Azerbaijan)